HW1 – CS 671

Yash Dagade

September 18, 2025

(a) No Regularization

1. **Why optimizing plain 0–1 loss is bad.** Because without a sparsity/complexity penalty a tree can grow arbitrarily deep to perfectly memorize the training set (zero training error) and badly overfit. In this case we would get really deep trees that may not generalize very well at all and would have just learning to memorize the training data and that's no good.

2. **Deepest valid tree.** Along any root-to-leaf path you can test each feature at most once, so I can have at most k splits. Since depth counts nodes (leaf included) and a tree with one leaf has depth 1, the deepest valid tree has depth

$$\boxed{k+1}.$$

3. **Number of valid optimal trees for Table 1.** Dataset: $(1,0) \to 0$ (twice), $(0,1) \to 1$ (once). Zero training error is achievable.

   - Depth-2 stumps:
     (a) Root split on x_1: $x_1{=}1 \mapsto 0$, $x_1{=}0 \mapsto 1$.
     (b) Root split on x_2: $x_2{=}1 \mapsto 1$, $x_2{=}0 \mapsto 0$.

     $\Rightarrow$ 2 trees.
   - Depth-3 extensions (root x_1, mirror for x_2):
     - Split only one child on x_2: 2 choices of child $\times 2$ free labelings $= 4$ trees.
     - Split both children on x_2: 2 leaves fixed, 2 free $\Rightarrow 2^2 = 4$ trees.

     $\Rightarrow$ 8 trees per root. Since root can be x_1 or x_2, total 16.

   Total valid optimal trees:

$$\boxed{18}.$$

4. **Unique logical expressions among these trees.** Constraints: $f(1,0) = 0$, $f(0,1) = 1$. Unseen points $(0,0)$ and $(1,1)$ are free, thus can be anything. That yields $2 \times 2 = 4$

distinct Boolean functions that produce the right answer for our constraints.

$$\begin{aligned}
&(0,0) \to 0,\ (1,1) \to 0 : f = \neg x_1 \wedge x_2, \\
&(0,0) \to 0,\ (1,1) \to 1 : f = x_2, \\
&(0,0) \to 1,\ (1,1) \to 0 : f = \neg x_1, \\
&(0,0) \to 1,\ (1,1) \to 1 : f = \neg x_1 \vee x_2.
\end{aligned}$$

So the number is

$$\boxed{4}.$$

(b) Soft Regularization

We define

$$R(f, X, y|\lambda) = L(f, X, y) + \lambda s(f),$$

where $s(f)$ is the number of leaves.

1. **Largest possible R for one-leaf tree.**

   A one-leaf tree (constant predictor) always predicts the majority class, so its error is at most $L = \frac{1}{2}$. Since it has $s(f) = 1$ leaf, we obtain

   $$R(f, X, y|\lambda) = L(f, X, y) + \lambda s(f) \ \le \ \tfrac{1}{2} + \lambda.$$

   Formally, this inequality is the correct upper bound for any $\lambda \ge 0$. However, if $\lambda > \frac{1}{2}$, the regularization penalty dominates: adding more leaves can never reduce the error by more than 1, so the trivial one-leaf tree will always be optimal. In that sense, while λ is not mathematically capped at $\frac{1}{2}$, values much larger than $\frac{1}{2}$ effectively force the trivial solution. Thus we can say

   $$R \ \le \ \tfrac{1}{2} + \lambda,$$

   or, if we restrict attention to the most meaningful range of λ,

   $$R \ \le \ \tfrac{1}{2} + \tfrac{1}{2} = 1.$$

2. **Accuracy needed for deeper prefix-refinement.** Refining f_d into f_{d+k} adds at least k leaves, so $s(f_{d+k}) \ge s(f_d) + k$. For R to improve:

   $$L(f_{d+k}) + \lambda(s(f_d) + k) < L(f_d) + \lambda s(f_d),$$

   hence

   $$L(f_{d+k}) \le L(f_d) - \lambda k.$$

   So the deeper tree must reduce misclassification by at least

   $$\boxed{\lambda k}.$$

3. **Depth guarantee for suboptimality.** Any depth-d tree has at least $s \geq d$ leaves, hence $R \geq \lambda d$. A depth-1 tree has $R \leq \frac{1}{2} + \lambda$. Therefore any depth d with

$$\lambda d > \tfrac{1}{2} + \lambda \quad \Longleftrightarrow \quad d > 1 + \tfrac{1}{2\lambda}$$

is always worse than some one-leaf tree. Thus the smallest such depth is

$$\boxed{d^*(\lambda) = \left\lfloor 1 + \tfrac{1}{2\lambda} \right\rfloor + 1}.$$

(If this exceeds the maximum feasible depth $k+1$, the statement is vacuously true.)

4. **Number of optimal trees for Table 1, $0 < \lambda < \frac{1}{2}$.**

Candidate objectives:

- One-leaf majority (predict 0): $R = \frac{1}{3} + \lambda$.
- Zero-loss stumps (2 leaves): $R = 2\lambda$.
- Deeper zero-loss trees (3 or 4 leaves): $R = 3\lambda$ or 4λ (never optimal for $\lambda > 0$).

Compare 2λ vs $\frac{1}{3} + \lambda$:

- $0 < \lambda < \frac{1}{3}$: stumps win $\Rightarrow$ $\boxed{2}$ optimal trees.
- $\lambda = \frac{1}{3}$: tie $\Rightarrow$ 2 stumps + 1 one-leaf $\Rightarrow$ $\boxed{3}$ optimal trees.
- $\frac{1}{3} < \lambda < \frac{1}{2}$: one-leaf wins $\Rightarrow$ $\boxed{1}$ optimal tree.